AI IN SDLC Assignment

What this assignment is

You will take one feature from a written specification through to a reviewed pull request on the workshop demo repository, using an AI agent to do the work while you hold the checks. The feature itself is small and could be hand-written in an afternoon. That is deliberate. What is being asked for is the evidence around it: a context file that earns its place, tests that would catch a real mistake, and a judgement call on every review finding.

Though the assignment is not tied to one vendor. But we suggest going for a free tier of GitHub copilot rather than a paid AI tool. Alternatives are welcome as long as it meet's compliance.

The feature to build

Add **approval limits by role** to the invoice approval service in `ai-sdlc-demo-app`:

- A Manager may approve invoices up to 50,000.
- A Director may approve invoices up to 500,000.
- Anything above 500,000 needs two distinct approvers, and the second approver must be a director.
- The same user may never count as both approvers on one invoice.
- A rejected attempt returns a clear reason, and nothing about the invoice changes.

The last two rules are where an agent is most likely to produce something that looks right and is not. Expect that and plan your tests around it.

Note: Use https://github.com/NITIN1670/ai-sdlc-demo-app.git repo for reference. There is no language constraint, so you are free to use any language for this assignment.

What to do

1. **Set up the context, then show that it matters.** Fork the repository and extend AGENTS.md so an agent arriving cold knows the approval rules, the layering, the test command and your review expectations.
2. **Use appropriate skills for the agents.**
3. **Use appropriate hooks for generating the structured and same format output.**
4. **Write the specification with a spec-driven tool.** Use Spec Kit, OpenSpec or BMAD, whichever you prefer, to produce the specification and task breakdown. Commit the files it generates.
5. **Implement it in at least two steps.** Do not ask for the whole feature in a single prompt. Land the single-approver limits first, with tests, then the two-approver rule. Each step is its own commit, with a message that says what changed and why.
6. **Make the tests mean something.**
    1. Write the unit tests for the approval rules and review every generated test.

    2. Add one Playwright end-to-end test that covers a rejected over-limit approval through the user interface.

7. **Review your own pull request with the agent, then judge the review.** Open the pull request, run the review prompt from the workshop against the diff, and in the description list every finding with one of three verdicts: fixed, dismissed with a reason, or deferred with a reason
8. **Log all AI interactions in a log file** - utilizing the capability of skills/hooks.
9. Create multiple agents and sub-agents for the assignment.

Install note: mutation testing with Stryker on .NET 8 needs an explicit version. Use `dotnet tool install -g dotnet-stryker --version 4.13.0`. The newest release targets a later runtime and will fail to install.

The demo video

Record one demo video of 8 to 10 minutes. Something must run live on screen: a test, the mutation run, the end-to-end test. A read-through of files is not a demo. Record with Teams by starting a meeting with only yourself and pressing Record, or use any screen recorder that produces an MP4

What to hand in

1. A pull request on your fork, with at least three commits that each stand on their own
2. The updated AGENTS.md
3. The specification files from whichever spec-driven tool you used
4. Unit tests for every approval rule, including the two negative rules
5. One Playwright test covering a rejected over-limit approval
6. Mutation scores before and after, pasted as text in the write-up
7. A pull request description listing every review finding with a verdict
8. A log file capturing all the AI interaction
9. ASSIGNMENT.md at the repository root, holding the write-up and the video link
10. The 8-to-10-minute demo video, uploaded and linked